BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 073 · SQL: SELECT, FROM and WHERE

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why close a file after use?

Q2. Which link best suits Bluetooth?
A. Nearby wireless keyboard
B. School buildings far apart
C. Undersea Internet cable
D. Long-distance fibre backbone

Q3. “Brute force works only by physically breaking the computer.” Correct this. Explain why.

Q4. A network shows 2001:db8::7. “IPv6 uses four decimal numbers separated by dots.” Correct this. How is this address written?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

SELECT — Specifies columns required from a database table.

FROM — Identifies the database table from which records are selected.

WHERE — Filters records using a condition.

SQL query — SELECT Name FROM Pupils WHERE Year = 10 retrieves relevant names from matching rows.

Algorithm — A defined set of processing instructions.

Testing — Checking expected outcomes against actual program behaviour.

New learning

SELECT: Specifies columns required from a database table. In the school club database, identify where this technique applies.

FROM: Identifies the database table from which records are selected. In the library books database, identify where this technique applies.

WHERE: Filters records using a condition. In the sports team records, identify where this technique applies.

SQL query: SELECT Name FROM Pupils WHERE Year = 10 retrieves relevant names from matching rows. In the museum member list, identify where this technique applies.

Worked example

A school club database uses select and from. Explain what each contributes. [4 marks]

1. Identify the program's purpose and the pieces of data needed.

2. SELECT [1]; Specifies columns required from a database table. [1].

3. FROM [1]; Identifies the database table from which records are selected. [1].

4. Each feature needs a correct technical fact plus an explanation.

We Do · Practise with guidance

Guided 1

A library books database needs select. Explain the purpose. [2 marks]

Worked method: SELECT [1]; Specifies columns required from a database table. [1].

Guided 2

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Explain from for a sports team records. [2 marks]

Worked method: FROM [1]; Identifies the database table from which records are selected. [1].

Guided 3

When programming a museum member list, explain why where and sql query matter. [4 marks]

Worked method: WHERE [1]; Filters records using a condition. [1]. SQL query [1]; SELECT Name FROM Pupils WHERE Year = 10 retrieves relevant names from matching rows. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete this SQL query to return names of pupils with a mark greater than 60.

```
SELECT Name FROM Pupils WHERE Mark ... 60
```

write

Question 1

State TWO relevant ideas or operations from this topic. [2 marks] Source: Original OCR J277-style teaching question; not an official past paper.

Question 2

Explain from and where in a café inventory. [4 marks] Source: Original OCR J277-style teaching question; not an official past paper.

Question 3

A programmer makes a school house records. Explain the roles of select, where and sql query. [6 marks] Source: Original OCR J277-style teaching question; not an official past paper.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

The SQL SELECT clause specifies which columns are retrieved from a database table.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It releases the handle or resources and helps ensure buffered writes are completed.

Q2. A — Nearby wireless keyboard

Q3. It tries many possible passwords or keys until one works.

Q4. IPv6 uses hexadecimal groups separated by colons. Groups of zeros can be shortened. Routers use IP addresses between networks; switches use MAC addresses within a LAN.

I Do / We Do pages · Guided practice

Q1. SELECT [1]; Specifies columns required from a database table. [1].

Q2. FROM [1]; Identifies the database table from which records are selected. [1].

Q3. WHERE [1]; Filters records using a condition. [1]. SQL query [1]; SELECT Name FROM Pupils WHERE Year = 10 retrieves relevant names from matching rows. [1].

You Do pages · Code First

```
SELECT Name FROM Pupils WHERE Mark > 60
```

Expected result: Ada, Kai (example Pupils rows: Ada 62; Ben 45; Kai 80)

You Do pages · Independent questions

Q1. SELECT [1]; FROM [1].

Q2. FROM [1]; Identifies the database table from which records are selected. [1]. WHERE [1]; Filters records using a condition. [1].

Q3. SELECT [1]; Specifies columns required from a database table. [1]. WHERE [1]; Filters records using a condition. [1]. SQL query [1]; SELECT Name FROM Pupils WHERE Year = 10 retrieves relevant names from matching rows. [1].

Exit Ticket · Model answer

True

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: